

AYMANE AIT DADS

Research Intern - Image & Video Processing | Deep Learning · Computer Vision · Numerical Optimization
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PROFESSIONAL SUMMARY

Ranked 1st in the EURECOM class leaderboard (0.791 AUC) on AI-generated image detection using CLIP ViT-L/14 with a custom ensemble and 10-view TTA, and submitted results to the NTIRE 2026 @ CVPR international benchmark. Achieved 99.8% accuracy and ROC AUC of 1.0 on aerial image classification across 17,500+ drone images, benchmarking CNN, DenseNet121, and hybrid CNN-Transformer architectures. Brings hands-on experience in deep learning, image processing, and computer vision alongside a proven track record of achieving significant results in competitive research settings. Aims to contribute to image, video, and geometry processing research at Disney Research Studios, applying rigorous ablation methodology and rapid problem-solving to production-quality research challenges.

CORE COMPETENCIES

Deep Learning | Computer Vision | Image Processing | Numerical Optimization | Software Engineering | Research & Development | Machine Learning | Quick Problem-Solving

WORK EXPERIENCE

Orange Maroc Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Implemented Random Forest and K-Means machine learning models on **100,000+ fiber-optic network records** to classify network quality across thousands of nodes using upload speed, latency, and jitter features.
- Developed an end-to-end clustering pipeline mapping customer performance profiles to **5 commercial service tiers**, with output adopted directly by the network optimization team.
- Reduced manual analysis time by **~60%** through automated feature engineering, model evaluation workflows, and structured reporting pipelines built in Python and Scikit-learn.
- Delivered stakeholder presentations translating **model outputs into actionable maintenance recommendations**, visualized via Power BI dashboards for non-technical audiences.

PROJECTS

AI-Generated Image Detection EURECOM ImSecu + NTIRE 2026 @ CVPR · 2025 · Ranked 1st EURECOM Class

- Ranked **1st in the EURECOM class leaderboard** (0.791 AUC) by fine-tuning CLIP ViT-L/14 (428M parameters) with a custom MLP head and dual learning-rate strategy across 250K images from 25+ generator types.
- Discovered that **1-epoch training (0.793 AUC) outperforms 4-epoch (0.767 AUC)** for out-of-distribution generalization; applied 3-model ensemble weighted by validation AUC and 10-view TTA, then submitted to NTIRE 2026 @ CVPR CodaBench international benchmark.

PyTorch · OpenCLIP · ViT · FP16 Mixed Precision · Kaggle GPU

Aerial Cactus Detection EURECOM Computer Vision · 2025 · 99.8% Accuracy

- Benchmarked CNN, DenseNet121, and hybrid CNN-Transformer architectures on **17,500+ drone images** for endangered species classification, achieving 99.8% accuracy, F1 = 0.999, and ROC AUC = 1.0.
- Designed and evaluated image classification pipelines with rigorous **numerical optimization and model selection**, comparing deep learning architectures across precision, recall, and AUC metrics.

PyTorch · DenseNet121 · CNN · Transformer · Scikit-learn

NVIDIA Nemotron Model Reasoning Challenge Kaggle · In Progress (June 2026) · \$106K+ Prize Pool

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, \$106K+ prize pool) in the NVIDIA Nemotron Reasoning Challenge by fine-tuning a **30B-parameter** Mamba-Transformer with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context).
- Performed **task-level error analysis** across 5 reasoning categories; built diagnostic evaluation pipeline revealing that over-expanded reasoning - not incorrect reasoning - was the primary failure mode across cipher and bit-manipulation tasks.

PyTorch · Unsloth · PEFT · vLLM · Hugging Face Transformers

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, Image Security, NLP, Statistics, Cloud Computing

CPGE - Mathematics & Physics - Ibn Timiya 2021 - 2023

SKILLS

Computer Vision & Image Processing: PyTorch | CLIP ViT | CNN | DenseNet | OpenCLIP | Image Classification | FP16/BF16 Mixed Precision

Machine Learning & Deep Learning: Scikit-learn | TensorFlow | LoRA / PEFT | Hugging Face Transformers | Ablation Studies | Ensemble Methods

Programming & Tools: Python | SQL | Bash | Git | Linux | Kaggle GPU | Docker (basic)